

Carola Bonamico

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Education

Politecnico di Torino, MSc in Computer Engineering – Software Sep 2024 – Present

- Weighted average: 27.8/30
- Main courses: Data Science and Database Technology, Software Engineering I/II, Web Applications I, Large Language Models for Software Engineering, Human Computer Interaction, Computer Architectures, Information Systems, System and Device Programming

University of Pavia, BSc in Biomedical Engineering Sep 2020 – Apr 2024

- Thesis: “Literature review on the impact of sleep quality on glycemic patterns in patients with type 1 diabetes”
- Main courses: Biomedical Data Processing, Models of Biological Systems

I.I.S. Cellini – Liceo Linguistico Alberti, Secondary School Diploma in Modern Languages Sep 2015 – Jun 2020

- Languages studied: English, French, German

Experience

Master's Thesis Student, ETH Zurich – Zurich, CH Mar 2026 – Present

- Carrying out my Master's thesis project at the Integrated Systems Laboratory (IIS), focused on developing Deep Learning models for silent speech recognition from electromyographic signals.

Software Developer, Squadra Corse PoliTo – Turin, IT Aug 2025 – Nov 2025

- Built a TUI in Go to send, receive and inspect CAN messages over the SocketCAN interface, streamlining on-track diagnostics and debugging.

Academic Tutor, University of Pavia – Pavia, IT Sep 2022 – Oct 2023

- Tutored small groups of students on the Fundamentals of Computer Science course and gave targeted one-to-one support on C programming exercises, covering algorithm design and memory management.

Technical Skills

Programming languages: C, Rust, Go, JavaScript, TypeScript, Python, ARM Assembly, MATLAB, ~~TeX~~

Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn

Libraries and frameworks: Node.js, React.js, React Native, TypeORM, Diesel

Databases: Oracle SQL, MongoDB, MySQL, SQLite, PostgreSQL

Tools and software: Git, GitHub, GitLab, Docker, Jupyter Notebook, LabVIEW

Other: REST APIs, WebSocket, JSON, Repository Pattern, Agile, UML, HTML, CSS

Personal Projects

CAN debug TUI

CAN bus debugging TUI written in Go, sending and receiving CAN messages over the SocketCAN interface with per-message control. Supports DBC files for message decoding.

Projects at ETH Zurich

SilentWear

Fork of SilentWear, a wearable system based on a textile neckband for silent speech recognition from EMG. Extended the existing CNN with a BiLSTM and a Transformer, with CTC decoding and beam search for direct EMG-to-text prediction. Integrated a dataset collected from 7 subjects, roughly 38 hours of recording, with isolated words and full sentences in both vocalised and silent conditions.

BioGUI

Contributor to BioGUI, an open source platform for real-time biosignal acquisition and labelling. Extended the data collection pipeline for a custom vocabulary protocol and implemented intra-file alignment across EMG, EEG and microphone signals in the binary .bio format, as well as inter-file alignment across recordings.

Projects at Politecnico di Torino

Participium

Full-stack platform for civic issue reporting in TypeScript/JavaScript (Node.js and web frontend), with role-based flows, geolocation and photo upload. Includes Telegram notifications and SonarQube for static analysis and continuous code quality monitoring.

Rust Chat

Real-time group messaging application built with Rust and React. Supports private chats, groups, JWT authentication, WebSocket communication and Docker deployment.

Geocontrol

Dockerised software monitoring system for managing geological, meteorological and environmental sensors. Built in TypeScript using the repository pattern, TypeORM and Node.js.

Pac-Man on LandTiger

Pac-Man game for the LandTiger LPC1768 board (ARM Cortex-M3), with an A* algorithm for ghost movement, a pause button and in-game audio through a speaker. Written in C and ARM assembly.

Certifications

Goethe-Zertifikat B1 (German), Goethe-Institut Italia

Issued: Jul 2019

B1 Preliminary (English), Cambridge English

Issued: May 2018